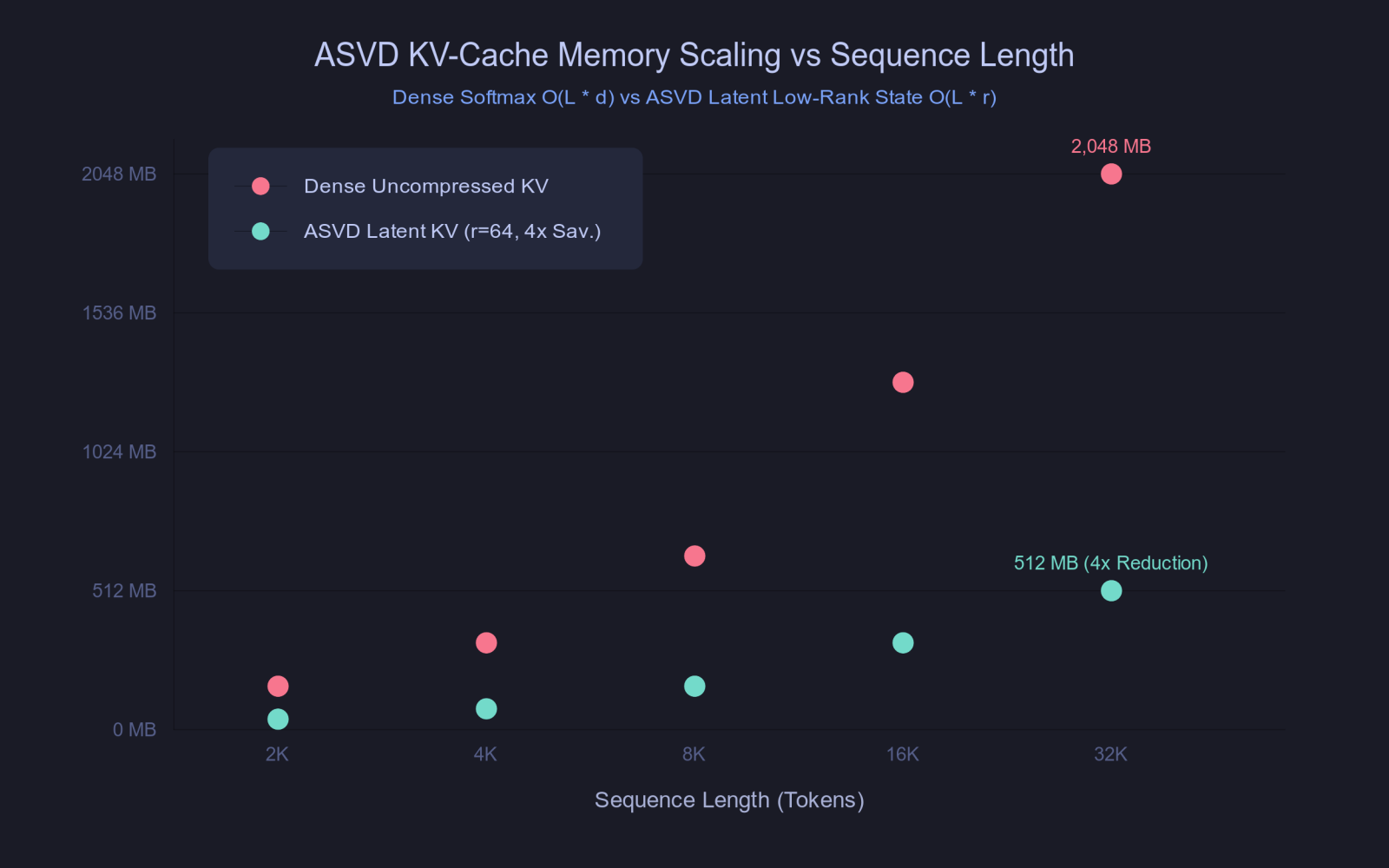

ASVD KV-Cache Memory Scaling vs Sequence Length
Dense Softmax O(L * d) vs ASVD Latent Low-Rank State O(L * r)
Dense Uncompressed KV
ASVD Latent KV (r=64, 4x Sav.)
2,048 MB
512 MB (4x Reduction)
2048 MB
1536 MB
1024 MB
512 MB
0 MB
2K
4K
8K
16K
32K
Sequence Length (Tokens)